TECHNICAL SUPPLEMENT 05 · ENGLISH

TOPSIS and Full-Space Pareto Analysis

Agri-Social Navigation in Vineyard Corridors

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Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Decision model

Three run-level criteria: mission utility UM, saturated clearance utility US and personal exposure fraction fP. UM=max(0,1-T/180) for successful missions and UM=0 for observed mission failures. US=0 for dmin≤0.45 m, US=(dmin-0.45)/0.75 for 0.45<dmin<1.20 m, and US=1 beyond 1.20 m. fP=E1.2/Tobserved is a cost. Utilities are transformed per run before averaging; minimum distance is not an unbounded benefit.

Vector normalization and explicit weights yield distances D+ and D- to the ideal and anti-ideal; closeness C=D-/(D++D-). The 180 s scale is an author-defined decision reference: mission timeout is wall-clock based and measured T is simulation time. Acceleration, jerk and stop counts are excluded.

Profile	w mission	w clearance	w exposure
equal	0.3333	0.3333	0.3333
group_balanced	0.5000	0.2500	0.2500
mission_priority	0.7000	0.1500	0.1500
safety_priority	0.2000	0.4000	0.4000

POOLED

Method	UM	US	fP	C equal	R eq	R group	R mission	R safety
M0	0.4669	0.4613	0.2728	0.6852	7	5	3	7
M1	0.0000	0.8435	0.8204	0.1576	9	9	9	9
M2	0.0000	0.8520	0.8192	0.1607	8	8	8	8
M3	0.2748	0.9040	0.0418	0.7882	6	7	7	5
M4	0.3451	0.8875	0.0735	0.8424	3	4	5	3
B1	0.3133	0.8605	0.0659	0.8126	5	6	6	4
B2	0.4257	0.7640	0.1646	0.8155	4	2	2	6
B3	0.4318	0.8542	0.0528	0.9210	1	1	1	2
B4	0.3727	0.9612	0.0069	0.8947	2	3	4	1

Dominance matrix: row dominates column (1=yes; 0=no), tolerance 1e-5.

	M0	M1	M2	M3	M4	B1	B2	B3	B4
M0	0	0	0	0	0	0	0	0	0
M1	0	0	0	0	0	0	0	0	0
M2	0	1	0	0	0	0	0	0	0
M3	0	1	1	0	0	0	0	0	0
M4	0	1	1	0	0	0	0	0	0
B1	0	1	1	0	0	0	0	0	0
B2	0	0	0	0	0	0	0	0	0
B3	0	1	1	0	0	0	1	0	0
B4	0	1	1	1	1	1	0	0	0

Spearman correlations over the nine method means:

	MissionUtility	SafetyUtility	PersonalExposureFraction
MissionUtility	1.000	-0.243	-0.293

	MissionUtility	SafetyUtility	PersonalExposureFraction
SafetyUtility	-0.243	1.000	-0.800
PersonalExposureFraction	-0.293	-0.800	1.000

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Method	UM	US	fP	C equal	R eq	R group	R mission	R safety
M0	0.4668	0.3931	0.3671	0.6042	7	7	5	7
M1	0.0000	0.8402	0.8645	0.1826	8	8	8	8
M2	0.0000	0.8388	0.8630	0.1823	9	9	9	9
M3	0.3222	0.8967	0.0492	0.8368	6	6	7	5
M4	0.3858	0.8870	0.0646	0.8816	4	5	6	4
B1	0.4058	0.8845	0.0384	0.9079	3	4	4	2
B2	0.4478	0.7797	0.0746	0.8758	5	3	3	6
B3	0.4467	0.8672	0.0495	0.9195	2	2	2	3
B4	0.4435	1.0000	0.0000	0.9750	1	1	1	1

Dominance matrix: row dominates column (1=yes; 0=no), tolerance 1e-5.

	M0	M1	M2	M3	M4	B1	B2	B3	B4
M0	0	0	0	0	0	0	0	0	0
M1	0	0	0	0	0	0	0	0	0
M2	0	0	0	0	0	0	0	0	0
M3	0	1	1	0	0	0	0	0	0
M4	0	1	1	0	0	0	0	0	0
B1	0	1	1	0	0	0	0	0	0
B2	0	0	0	0	0	0	0	0	0
B3	0	1	1	0	0	0	0	0	0
B4	0	1	1	1	1	1	0	0	0

Spearman correlations over the nine method means:

	MissionUtility	SafetyUtility	PersonalExposureFraction
MissionUtility	1.000	-0.310	-0.259
SafetyUtility	-0.310	1.000	-0.767
PersonalExposureFraction	-0.259	-0.767	1.000

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Method	UM	US	fP	C equal	R eq	R group	R mission	R safety
M0	0.4671	0.8355	0.1883	0.8142	7	4	3	7
M1	0.0000	0.8498	0.8253	0.0084	8	8	8	8
M2	0.0000	0.8483	0.8254	0.0077	9	9	9	9
M3	0.3700	0.9130	0.0316	0.8916	3	7	7	3

Method	UM	US	fP	C equal	R eq	R group	R mission	R safety
M4	0.3852	0.8884	0.0742	0.8841	4	5	5	5
B1	0.3766	0.8745	0.0626	0.8822	5	6	6	4
B2	0.4508	0.8320	0.1537	0.8436	6	3	2	6
B3	0.4461	0.8704	0.0481	0.9432	2	1	1	2
B4	0.4178	0.9379	0.0086	0.9460	1	2	4	1

Dominance matrix: row dominates column (1=yes; 0=no), tolerance 1e-5.

	M0	M1	M2	M3	M4	B1	B2	B3	B4
M0	0	0	0	0	0	0	0	0	0
M1	0	0	1	0	0	0	0	0	0
M2	0	0	0	0	0	0	0	0	0
M3	0	1	1	0	0	0	0	0	0
M4	0	1	1	0	0	0	0	0	0
B1	0	1	1	0	0	0	0	0	0
B2	0	0	0	0	0	0	0	0	0
B3	0	1	1	0	0	0	0	0	0
B4	0	1	1	1	1	1	0	0	0

Spearman correlations over the nine method means:

	MissionUtility	SafetyUtility	PersonalExposureFraction
MissionUtility	1.000	-0.276	-0.243
SafetyUtility	-0.276	1.000	-0.783
PersonalExposureFraction	-0.243	-0.783	1.000

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Method	UM	US	fP	C equal	R eq	R group	R mission	R safety
M0	0.4668	0.2515	0.3396	0.5900	7	7	5	7
M1	0.0000	0.8364	0.7963	0.2299	9	9	9	9
M2	0.0000	0.8790	0.7972	0.2426	8	8	8	8
M3	0.3551	0.9078	0.0389	0.8771	5	6	7	3
M4	0.3863	0.8903	0.0617	0.8929	4	4	4	5
B1	0.3834	0.8762	0.0534	0.8930	3	5	6	4
B2	0.4320	0.7866	0.1517	0.8305	6	3	3	6
B3	0.4446	0.8473	0.0483	0.9312	2	2	1	2
B4	0.4382	0.9442	0.0033	0.9695	1	1	2	1

Dominance matrix: row dominates column (1=yes; 0=no), tolerance 1e-5.

	M0	M1	M2	M3	M4	B1	B2	B3	B4
M0	0	0	0	0	0	0	0	0	0
M1	0	0	0	0	0	0	0	0	0

	M0	M1	M2	M3	M4	B1	B2	B3	B4
M2	0	0	0	0	0	0	0	0	0
M3	0	1	1	0	0	0	0	0	0
M4	0	1	1	0	0	0	0	0	0
B1	0	1	0	0	0	0	0	0	0
B2	0	0	0	0	0	0	0	0	0
B3	0	1	0	0	0	0	1	0	0
B4	0	1	1	1	1	1	1	0	0

Spearman correlations over the nine method means:

	MissionUtility	SafetyUtility	PersonalExposureFraction
MissionUtility	1.000	-0.276	-0.377
SafetyUtility	-0.276	1.000	-0.600
PersonalExposureFraction	-0.377	-0.600	1.000

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Method	UM	US	fP	C equal	R eq	R group	R mission	R safety
M0	0.4669	0.3649	0.1964	0.7299	2	2	1	6
M1	0.0000	0.8478	0.7953	0.1828	8	8	8	8
M2	0.0000	0.8421	0.7912	0.1815	9	9	9	9
M3	0.0521	0.8985	0.0473	0.5571	6	7	7	4
M4	0.2233	0.8842	0.0935	0.6828	4	4	4	3
B1	0.0873	0.8066	0.1095	0.5533	7	6	6	5
B2	0.3721	0.6577	0.2784	0.6996	3	3	3	7
B3	0.3897	0.8320	0.0653	0.8714	1	1	2	1
B4	0.1913	0.9629	0.0157	0.6777	5	5	5	2

Dominance matrix: row dominates column (1=yes; 0=no), tolerance 1e-5.

	M0	M1	M2	M3	M4	B1	B2	B3	B4
M0	0	0	0	0	0	0	0	0	0
M1	0	0	0	0	0	0	0	0	0
M2	0	0	0	0	0	0	0	0	0
M3	0	1	1	0	0	0	0	0	0
M4	0	1	1	0	0	1	0	0	0
B1	0	0	0	0	0	0	0	0	0
B2	0	0	0	0	0	0	0	0	0
B3	0	0	0	0	0	1	1	0	0
B4	0	1	1	1	0	1	0	0	0

Spearman correlations over the nine method means:

	MissionUtility	SafetyUtility	PersonalExposureFraction
MissionUtility	1.000	-0.510	-0.293
SafetyUtility	-0.510	1.000	-0.517
PersonalExposureFraction	-0.293	-0.517	1.000

The pooled equal-weight order starts B3, B4, M4. The pooled mean-space Pareto set is M0, B3, B4. M4 is not declared non-dominated on those pooled mean decision vectors. These are conditional rankings, not a demonstrated task optimum or field-safety ordering.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.